

Rational Points on Elliptic Curves Chapter 2

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March 2025

1 Notes

The intersection of the cubic with the x -axis along with $\mathcal{O}$ forms K_4 in $\mathbb{C}$. The group could be C_2 in $\mathbb{R}$ (if f has exactly one real root). Over $\mathbb{Q}$, G could be trivial, C_2 , or K_4 . The points of order 2 are completely described.

Suppose $3P = \mathcal{O}$, $P \neq \mathcal{O}$. Then $2P = -P$, so $x(2P) = x(-P) = x(P)$. If $x(2P) = x(P)$, then $2P = \pm P$, so either $P = \mathcal{O}$ or $3P = \mathcal{O}$, the former of which is excluded by assumption. The second also excludes $P = \mathcal{O}$, so if $x(2P) = x(P)$ and $P \neq \mathcal{O}$, then P is of order 3.

Points of order 3. Note that $\psi_3(x) = 2f(x)f''(x) - f'(x)^2$. $\psi'_3(x) = 12f(x)$. If α is a common root of ψ_3 and ψ'_3 , then α is a common root of f and f' , contradicting that C is non-singular. Let $\beta_1, \beta_2, \beta_3, \beta_4$ be the distinct complex roots of ψ and $\delta_i^2 = f(\beta_i)$. $\delta_i \neq 0$ as otherwise $(\beta_i, 0)$ would be of both order 2 and 3, contradiction. Hence, each δ_i is nonzero, and

$$\{(\beta_1, \pm\delta_1), (\beta_2, \pm\delta_2), (\beta_3, \pm\delta_3), (\beta_4, \pm\delta_4)\}$$

contains exactly eight points. These points along with $\mathcal{O}$ form a group G of 9 points isomorphic to $C_3 \times C_3$ as no element has order 9.

Real and Complex Points on Cubic Curves. Note

$$\{\mathcal{O}\} \subset C(\mathbb{Q}) \subset C(\mathbb{R}) \subset C(\mathbb{C}).$$

$C(\mathbb{R})$ is a one-dimensional Lie group. If $C(\mathbb{R})$ has one connected component, it is isomorphic to $\mathcal{U}$. The elements of order dividing m in $\mathcal{U}$ are

$$\{1, e^{2\pi i/m}, e^{4\pi i/m}, \dots, e^{2\pi(m-1)/m}\},$$

which is C_m . Hence the points of order m in $C(\mathbb{R})$ is C_m . If $C(\mathbb{R})$ has two connected components, the group is isomorphic to $\mathcal{U} \times C_2$. If m is odd, the group in $C(\mathbb{R})$ is still C_m . If m is even, the group is $C_m \times C_2$. In particular, for $m = 3$, the group is C_3 . As a consequence, ψ_3 at least two real roots, one of which $f(\beta_i) > 0$ and the others have $f(\beta_i) < 0$.

Lattice Points of Finite Order. The discriminant D is a linear combination of f and f' when f is monic.

Let $P, 2P$ be two lattice points. Then $y = 0$ or $y|D$.

Proof: If $y = 0$, we're done. Otherwise we can compute $x(2P) = \frac{f'(x)^2 - 4(a+2x)f(x)}{4f(x)} = \frac{f'(x)^2}{4f(x)} - (a+2x) \in \mathbb{Z}$. Hence $f(x) \mid f'(x)^2$. D is in the ideal generated by f, f' in $\mathbb{Z}[x]$, so there exist integer polynomials u, v for which $uf + vf' = D$. Now $f(x) \mid f'(x)^2 v(x)^2$, so $f(x) \mid (D - u(x)f(x))^2$, or $f(x) \mid D^2$. Since $f(x) = y^2$, we have $y \mid D$.

Points of Finite Order are Lattice Points. We can factor out p from x, y . Number theory shows that $\text{ord}(x) = 2\nu$ and $\text{ord}(y) = 3\nu$ for some integer ν . This motivates the definition $C(p^\nu) = \{(x, y) \in C(\mathbb{Q}) : \text{ord}(x) \leq -2\nu \text{ and } \text{ord}(y) \leq -3\nu\}$. We claim that $C(p^\nu) < C(\mathbb{Q})$. We proceed by finding formulas for $(t_1, s_1) + (t_2, s_2)$ and chasing the orders. We find $t(P_1) + t(P_2) - t(P_1 + P_2) \in p^{3\nu}R$ (for $t_1 \neq t_2$). We may write this as: there is a one-to-one homomorphism $C(p^\nu)/C(p^{3\nu}) \rightarrow (p^\nu R)/(p^{3\nu}R)$.

2(b). Let p be a prime dividing M and consider a Sylow p -group G_p such that $|G_p| = 2n$ for some $n > 0$. By the FTFGAG, $G_p = C_{p^{a_1}} \times C_{p^{a_2}} \times \cdots \times C_{p^{a_k}}$ for some $a_i > 0$. The number of elements with order dividing p is p^k . Since $p^k = p^2$, $k = 2$. Write $G_p = C_{p^a} \times C_{p^b}$ such that $0 < a \leq b$ and $a + b$ is even. The number of elements with order dividing $m = p^{(a+b)/2}$ is $p^{a+(a+b)/2}$. By givens, this count also equals $m^2 = p^{2(a+b)/2}$, so $a = b$.

By FTFGAG, $A = \prod G_p = \prod C_{p^j}^2 = (\prod C_{p^j})^2 = C_M^2$.

2. (a) Use

$$x(2P) = \frac{f'(x)^2 - 4(a+2x)f(x)}{4f(x)}.$$

Find that

$$\psi_3(x) = 2f''(x)f(x) - f'(x)^2.$$

4 (b). Conjecture: $M_p = p + 1$. Claim: the residues $x^3 + 1$ are distinct modulo p for distinct x . Suppose $x^3 \equiv y^3 \pmod{p}$ for $x, y \not\equiv 0$. Then $(xy^{-1})^3 \equiv 1 \pmod{p}$. Since $(xy^{-1})^{p-1} \equiv 1 \pmod{p}$ by FLT, we have $(xy^{-1})^{\gcd(3, p-1)} \equiv 1 \pmod{p}$. Hence, $xy^{-1} \equiv 1 \pmod{p}$ as $p-1 \equiv 1 \pmod{3}$, so $x \equiv y \pmod{p}$. x^3 is nonzero for nonzero x , so the residues are distinct. We can now replace $x^3 + 1$ with z in $\mathbb{F}_p$.

If $z = 0$, then $y = 0$. Else $z \neq 0$ and is a QR modulo p , each of which

produces two values of y . Hence there are $1 + 2(p-1)/2 = p$ values of M_p in $\mathbb{F}_p$. We include $\mathcal{O}$ so the count is $p+1$.

7(b). Consider the set $S = \{k + pR : k = 0, 1, \dots, p-1\}$ and let $x = \frac{m}{n}p^\nu \in R$ for m and n prime to p . If $\nu > 0$, then $x + pR = 0 + pR$. Else $x = m/n$. We want $x - k = \frac{m-nk}{n} \in pR$, or $k = mn^{-1} \pmod{p}$. Hence there exists a unique element of S that corresponds to x (that is, $x + pR = k + pR$). S has exactly p elements as otherwise $k - j \in pR$ for some integers $k \neq j$ in the set $\{0, 1, \dots, p-1\}$.

R/pR is a ring with p elements and is more specifically the field $\mathbb{F}_p$. Hence, pR is a maximal ideal.

(c). Assume a and b are coprime. If $v_p(a) = 0$, then a/b is a unit. Else $v_p(a) > 0$, so $a/b \in pR$.

(e). Let $I \subset R$. If $I \neq 0$, there exists a positive integer n such that $n = v_p(x)$ is minimal over $x \in I$. Then $I = p^n R$. It follows pR is the only maximal ideal as $pR \supset p^2 R \supset p^3 R \supset \dots \supset 0$.

10. $\mathcal{O}$ has order 1. Assume our point has rational coordinates. If $y = 0$, then $x = 0$. The point $(0, 0)$ lies on the x -axis and has order 2. Else assume $y > 0$. Further by strong Nagell-Lutz y is an integer such that y^2 divides the discriminant $-4p^3$ of the cubic. If $p = 2$, then $y^2 \mid 32$ or $y \mid 4$, so $y = 1, 2, 4$. We also have $x(x^2 + p) = y^2$, so $x \mid y^2$. Checking cases gives no solutions. Else $p > 2$ and $y = 1, 2, p, 2p$. Each case again gives no solutions. The only points of finite order on each curve are $\mathcal{O}$ and $(0, 0)$.

13. (a) $D = a^2 b^2 - 4b^3 = 0$. Then $b = 0$ or $4b = a^2$. $a = 1 - 2t$ and $b = t^2$, so $t = 0$ or $4t^2 = 4t^2 - 4t + 1$, or $t = 1/4$. C is singular iff $x \in \{0, 1/4\}$.

(b) Let $P = (t, t)$. Since $t \neq 0$, P doesn't lie on the x -axis, so P can't have order 2. Note that $f'(x) = 3x^2 - (4t - 2)x + t^2$, so $f'(t) = 2t$. $f(t) = y^2 = t^2$. Then

$$\begin{aligned} x(2P) &= \frac{f'(t)^2 - 4(a + 2t)f(t)}{4f(t)} \\ &= \frac{4t^2 - 4(1 - 2t + 2t)t^2}{4t^2} \\ &= 0, \end{aligned}$$

Note that $f(0) = y^2 = 0$, so $2P = (0, 0)$ and in particular lies on the x -axis. $2P$ has order 2, so P has order 4.

(c). $2P' \neq \mathcal{O}$ and $y(2P') = 0$. Shift in the x -direction C' so that $x(2P') = 0$. C' is now of the form $y^2 = x^3 + ax^2 + bx$. P' cannot lie on the x -axis, so we may map $x \rightarrow \lambda^2 x$ and $y \rightarrow \lambda^3 y$ so that our new P' has equal x and y coordi-

notes. In particular we pick $\lambda = x/y$. The condition $x(2P') = 0$ gives using the duplication formula $b = t^2$. $P' = (t, t)$ lies on C' , so $t^2 = t^3 + at^2 + t^3 + t^2$, or $a = 1 - 2t$. C' is now of the desired form.

(d). ?